Voltage: energy per charge

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret potential difference as energy transferred per unit charge.

SUCCESS CRITERIA - I can...

1. Use volts as joules per coulomb.
2. Calculate energy transfer with E=qV.
3. Distinguish voltage from current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “voltage” and “charge” to compare your model or result. Sentence starters are optional.

voltage: electric potential difference; energy transferred per charge

charge: a property responsible for electric interactions

energy: a quantity tracked through storage and transfers in a system

01 Warm-up: bring back an earlier idea

Two objects have charges -3 nC and -5 nC. **Predict** attraction or repulsion and draw both force directions.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: energy transferred = qV; V = energy transferred/q. Magnitudes here; 1 V = 1 J/C. Voltage is a difference between two points.

Worked reasoning: Moving 2 C through a potential difference of 3 V changes electric potential energy by q ΔV=6 J. Voltage is energy change per charge, not energy by itself.

Your turn: Draw two labeled potential levels and an energy account for a positive charge moved between them.

Sentence starter: The … in my model represents … . Its label shows … .

Initial / supply _____ V: ____ J/C

Final / component _____ V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Compare a 3 V and a 6 V battery for the same amount of charge transferred. Which supplies more energy per coulomb?

Sentence starter: Both … . They differ in … because … .

Initial / supply

_____

V: ____ J/C

Final / component

_____

V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

04 Apply the physics

A 6 V battery moves 2 C through a model-car circuit.
Find the energy transferred.
Label voltage as energy per coulomb and show the total transfer on the model.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial / supply

_____

V: ____ J/C

Final / component

_____

V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says a 9 V battery contains 9 coulombs.
Explain the units and why voltage does not state total battery capacity.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

Voltage: energy per charge

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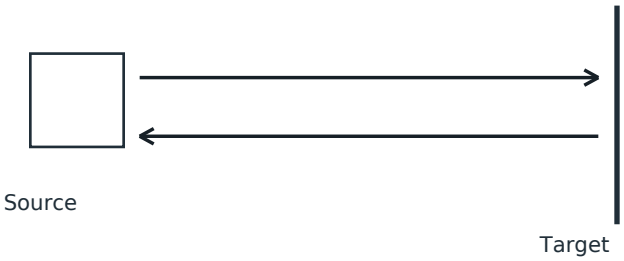
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A wall is 60 m away. An echo returns after 0.6 s. A student divides one-way distance by this time.
Find the correct speed and explain the student's error.
Label the full travel path and the recorded time.

Word help: echo = a sound received after reflection

Sentence starter: The step I would change is … . The evidence is … .



Marzano check - circle: 1 2 3

07 Exit check

What does a potential difference of 4 V mean in J/C?

Sentence starter: Both … . They differ in … because … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two devices use the same 6 V supply but move charges 1 and 3 C.
Compare transferred energies.

Sentence starter: Both … . They differ in … because … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3